

Morphic Studio Product Architecture Specification

Document 57 — Visual Creation Studio, Image Generation & Design System Specification

Version: 1.0

Status: Approved

1. Purpose

This document defines the Visual Creation Studio and Image Generation System for Morphic Studio.

Its purpose is to provide a unified environment for concept art, character design, environments, storyboards, comic panels, mood boards, and other visual assets while ensuring complete integration with the platform's narrative intelligence, Story Bible, Character Studio, World Builder, and AI systems.

2. Design Philosophy

The Visual Studio follows these principles:

- Images belong to stories.
 - Visual canon is as important as narrative canon.
 - AI generates while respecting established identities.
 - Human creators remain artistic directors.
 - Visual assets evolve through versioning.
 - Every generated image remains traceable.
 - Visual consistency is continuously maintained.
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3. Visual Workspace

The workspace consists of:

Navigation Panel

- Projects
- Characters

- Environments
- Storyboards
- Mood Boards
- Asset Collections

Generation Canvas

- Prompt editor
- Reference images
- Live previews
- Iteration history

Inspector Panel

- Metadata
- Generation settings
- Character bindings
- Style profiles
- Version history
- Canon validation

Asset Browser

- Generated images
- Imported references
- Previous versions
- Style libraries
- Templates

4. Image Generation

Support multiple generation workflows including:

- Text-to-Image
- Image-to-Image
- Sketch-to-Image
- Character Portrait Generation
- Environment Generation
- Creature Design
- Object Design
- Prop Design
- Comic Panel Generation
- Storyboard Generation

Generation passes through the Unified AI Gateway.

5. Character Design Integration

Images may reference Character Studio entities.

Automatically inherit:

- Appearance
- Clothing
- Hair
- Eye color
- Body proportions
- Age
- Species
- Expressions
- Injuries
- Accessories

Changes to approved visual canon update future generations.

6. Environment Design

Generate and manage:

- Cities
- Buildings
- Interiors
- Landscapes
- Vehicles
- Architecture
- Fantasy environments
- Science fiction environments

World Builder metadata informs visual generation.

7. Style Library

Maintain reusable visual styles.

Examples:

- Anime
- Realistic

- Semi-realistic
- Comic
- Manga
- Pixar-inspired
- Watercolor
- Oil painting
- Pixel art
- Studio-specific styles

Projects may define custom style profiles.

8. Prompt Intelligence

Prompts are enriched automatically using:

- Story Bible
- Character profiles
- World lore
- Canon rules
- Style preferences
- Timeline context
- Previous approved images

Users may edit the final prompt before generation.

9. Reference Management

Support:

- Reference images
- Pose references
- Color palettes
- Mood boards
- Composition guides
- Lighting references
- Costume references

References remain linked to related assets.

10. Editing Tools

Provide AI-assisted editing including:

- Inpainting
- Outpainting
- Background replacement
- Object removal
- Style transfer
- Expression changes
- Pose refinement
- Upscaling
- Color correction
- Cropping
- Aspect ratio conversion

Edits create new asset versions without overwriting originals.

11. Storyboards & Comics

Support visual storytelling workflows including:

- Scene planning
- Shot sequencing
- Comic panel layout
- Camera framing
- Dialogue placement
- Transition planning

Storyboard assets connect directly to scenes in the Story Workspace.

12. Visual Canon Engine

Maintain consistency across generated assets.

Validate:

- Character appearance
- Costume continuity
- World design
- Color language
- Architectural consistency

- Technology level
- Historical accuracy
- Timeline-dependent appearances

Potential inconsistencies are surfaced before approval.

13. Asset Versioning

Every visual asset maintains:

- Version history
- Generation parameters
- Prompt history
- AI model information
- Related project entities
- Approval status
- Creator attribution

Creators may compare, restore, or branch visual assets.

14. Collaboration

Support:

- Comments
- Reviews
- Approval workflows
- Team annotations
- Version comparison
- Art direction requests
- Task assignments

Visual feedback integrates with the Collaboration module.

15. Integration

Visual Studio integrates with:

- Character Studio
- World Builder
- Story Workspace
- Story Bible

- Canon Engine
- AI Gateway
- Knowledge Graph
- Asset Manager
- Workflow Engine
- Publishing Center

Generated assets automatically inherit relevant project context.

16. Analytics

Provide insights including:

- Generation volume
- Prompt effectiveness
- Style usage
- Asset reuse
- Canon consistency
- Approval rate
- AI model performance
- Generation costs

Analytics support creative optimization without constraining artistic freedom.

17. Acceptance Criteria

The Visual Creation Studio is successful if:

- Generated visuals remain consistent with project canon.
 - Character appearances remain recognizable across generations.
 - Style profiles maintain visual identity.
 - Storyboards connect directly to narrative structure.
 - Asset versioning protects creative work.
 - AI suggestions remain explainable and editable.
 - Visual assets integrate seamlessly with the broader Morphe Studio ecosystem.
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Conclusion

The Morphe Studio Visual Creation Studio transforms AI image generation into a structured visual storytelling system. By integrating image creation with the Story Bible, Character Studio, World Builder, Canon Engine, Knowledge Graph, and Asset Manager, the platform enables creators to develop coherent

visual identities for stories, characters, and worlds while preserving artistic control, canonical consistency, and long-term maintainability.